

Ayuvedic Disease Prediction and Recommendation System -Medicine Recommendation Module

Project ID: 24 - 25J – 305

Project Proposal Report

ROCHE J P-IT22344274

Supervised by Ms Manori Gamage

Co-supervised by Ms Uthpala Samarakoon

B.Sc. (Hons) Degree in Information Technology Specializing in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

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I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Roche J P	IT22344274	

The above candidate is carrying out research for the undergraduate Dissertation under my Supervision

Name of the Supervisor: Ms Manori Gamage

Name of the Co-Supervisor :Ms Uthpala Samarakoon

.....
Signature of the Supervisor

.....
Date

.....
Signature of the Co-supervisor

.....
Date

ABSTRACT

Ayurveda is one of the oldest holistic medical systems in the world. It offers valuable knowledge about herbs and their healing properties. Unfortunately, the lack of organized datasets limits its use in modern healthcare. This study suggests a machine learning-based herb recommendation system to digitize and personalize Ayurvedic knowledge. The research has three main stages ,First Stage is using Named Entity Recognition to identify herbs, diseases, doshas, and related factors from classical texts and reliable literature, Second stage is creating a structured dataset that connects diseases to herbs, including details like body constitution , severity, age group, and contraindications; and Third Stage training machine learning models, such as Random Forest, Support Vector Machine, and Decision Tree, to recommend the most appropriate herb(s) for a disease. The expected results include strong classification performance, with Random Forest likely achieving the highest accuracy. The system seeks to improve access to Ayurvedic herb knowledge while providing personalized, evidence-based recommendations.

Key terms – Ayurveda, Named Entity Recognition , Herb Recommendation, Random Forest

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
NER	Named Entity Recognition
SVM	Support Vector Machine
IM	Indigenous Medicine
CNN	Convolution Neural Networks

1 INTRODUCTION

The natural, age-old practice of Sri Lankan Ayurvedic medicine dates back more than 3,000 years. This distinctive style of Ayurveda, known locally as Hela Wedakama a term that symbolizes the medicinal techniques employed by the Sinhalese population[1], has its roots in ancient Vedic traditions and is influenced by the island's indigenous knowledge and Buddhist philosophy. It places an emphasis on balance, natural remedies, and individualized treatment plans. In Sri Lanka, Ayurveda is more than just a therapeutic approach; it is a way of life that affects daily routines, eating patterns, and general health. It primarily uses herbs as therapeutic agents and aims to treat the underlying causes of illnesses in order to restore balance between the mind, body, and spirit.

Through the Department of Ayurveda and the Ministry of Indigenous Medicine, which are in charge of practitioner certification, medication standardization, and research projects, the Sri Lankan government actively supports and regulates Ayurvedic medicine. Formal Ayurvedic degrees are offered by educational establishments such as the University of Colombo's Institute of Indigenous Medicine, and current research attempts to combine Ayurveda with contemporary medications and diagnostics. As a reflection of its incorporation into the national healthcare system, ayurvedic hospitals coexist with allopathic medical facilities.

Ayurvedic knowledge is still not well-organized, even though it has a long history. It is spread out across classical texts, which makes it hard to find and use in a systematic way. To tackle this issue, this study suggests a machine learning-driven Medicine recommendation system that digitizes and categorizes Ayurvedic knowledge to offer tailored, evidence-based medical recommendations. Named Entity Recognition will be used to pull out herbs, diseases, body constitutions and other things from classical texts. The information will be organized into a dataset that connects diseases to herbs. This dataset will include information like, severity, age group, and contraindications. Train machine learning models like Random Forest, Support Vector Machine and Decision Tree to suggest the best herb(s) for a specific disease. This approach bridges the gap between traditional healing and modern technology, making herbal medicine more accessible in today's digital era

Today, Ayurvedic medicine in Sri Lanka is a unique blend of traditional practices, cultural heritage, and biodiversity. Although it has deep historical roots, it still plays a key role in healthcare by providing both preventive and curative treatments. The wealth of this medical heritage, along with a large variety of medicinal plants, makes Sri Lanka an important center for the development and study of Ayurveda today.

1.1 Background & Literature survey

Background

Indigenous Medicine (IM), locally referred to as Helawedakama,[2] is an ancient heritage of Sri Lanka that has been preserved and practiced over the centuries generation after generation. Sri Lanka holds a unique place in the history of medicine, as it is believed to be the first country on the planet to have organized hospitals. Archaeological evidence from the excavations in old cities such as Polonnaruwa, Medirigiriya, Anuradhapura, and Mihinthale reveals the remains of ancient medical facilities, which are considered to be one of the world's oldest hospitals.

Native physicians enjoyed a very high status in Sri Lankan society from time immemorial due to the royal patronage they enjoyed from ancient kings. This not only improved their social standing but also facilitated the preservation and advancement of native medical knowledge.

Literature Review

Ontology-Based Personalized Herb Recommendation

To enhance semantic reasoning and personalized recommendations in traditional medicine, Promkot et al. proposed a Personalized Traditional Medicine Recommendation System using ontology modeling and rule inference [3]. Their system, though focused on Thai traditional medicine demonstrates key concepts that can be translated to Ayurveda Such as mapping symptoms, herbal properties, and diseases using ontological structures. Through a rule-based inference engine, the system provides tailored herb recommendations by aligning user-specific conditions with herbal attributes and treatment goals. This approach shows the value of semantic representation in organizing complex traditional knowledge systems for computational processing

Data-Driven Modeling in Traditional Medicine

Further supporting data-driven herb recommendation, Zhao et al. [4] introduced a Clinical Herb Modeling and Herb Pair Recommendation System in the context of Traditional Chinese Medicine (TCM). Their system utilized machine learning algorithms on a large-scale clinical dataset to identify frequent herb combinations (herb pairs) and recommend them based on disease patterns. The model incorporated feature extraction, herb-pair co-occurrence analysis, and deep learning techniques to predict effective herbal formulations. While developed for TCM, their methodology showcases a scalable framework for building intelligent herb recommendation systems, which can be adapted for Ayurvedic practice.

Predicting Ayurveda-Based Constituent Balancing Using Machine Learning

In a significant contribution to this field, Vishu Madaan and Anjali Goyal [5] developed a machine learning-based system to predict imbalances in the Ayurvedic doshas Vata, Pitta, and Kapha based on Prakriti (bodily constitution) assessment. Their approach relied on a self-administered questionnaire designed in alignment with traditional Ayurvedic principles to collect physiological, physical, and psychological data from users. The data underwent preprocessing and was then analyzed using several classification algorithms, including Naive Bayes, Random Forest, and Decision Tree classifiers. Among these, the Random Forest algorithm achieved the highest prediction accuracy of 93.05%, demonstrating its effectiveness in managing the nonlinear and categorical nature of Ayurvedic inputs. The system offers a structured way to digitize and modernize Ayurvedic diagnostics, enabling personalized identification of dosha imbalances that can subsequently inform individualized herbal or lifestyle recommendations.

Personalized Ayurvedic Medicine Recommendations Using Decision Tree Algorithms

Vijay M et al. [6] presented a comprehensive Ayurvedic Medicine Recommendation System (AMRS) that integrates decision tree algorithms with symptom-based analysis and Google Maps integration for clinic localization at the 2024 International Conference on Emerging Research in Computational Science (ICERCS). Their system achieved impressive performance metrics with 95.34% accuracy on test data and 92.12% average accuracy using 10-fold cross-validation. The decision tree classifier

demonstrated high precision rates for specific conditions: Diarrhea (96%), Gastritis (92%), Arthritis (94%), and Migraine (98%), with corresponding recall values of 94%, 88%, 93%, and 96% respectively. The system's overall performance metrics showed 85% accuracy, 82% precision, 80% recall, and 81% F1-score in real-world testing scenarios.

The implementation incorporated both traditional Ayurvedic principles and modern technological solutions, utilizing an 80-20 training-testing data split for model development. The system's unique contribution lies in its dual functionality: providing personalized medicine recommendations based on user symptoms and demographic data while simultaneously offering location-based services through Google Maps integration to help users find nearby Ayurvedic clinics. User experience evaluation revealed 90% positive feedback, with 85% of users particularly valuing the convenience of the integrated clinic locator feature. This approach successfully bridges the gap between traditional Ayurvedic practices and modern technology, demonstrating that ancient medical wisdom can be effectively digitized while maintaining its core therapeutic principles.

AyurVedaMitra

Varsha et al. (2025) introduced AyurVedaMitra, a technological innovation that demonstrates the integration of machine learning algorithms with traditional Ayurvedic medicine through mobile application development. The research employs Convolutional Neural Networks (CNN)[7] and Random Forest algorithms to enhance precision medicine approaches within the Ayurvedic healthcare framework, contributing to the growing body of literature that explores the digitization of traditional knowledge systems and medical services. This work exemplifies the current trend toward incorporating artificial intelligence in healthcare delivery, particularly in post-pandemic scenarios where digital health solutions have become increasingly vital. The platform addresses the technological gap in Ayurvedic practice by leveraging ensemble learning approaches that can handle complex medical datasets while maintaining interpretability, representing a significant advancement

1.2 Research gap

Although Ayurveda has been an integral part of Sri Lankan health care for centuries, its integration with modern-day digital technologies remains in a very elementary phase. Although a growing number of papers have debated the use of artificial intelligence (AI) and machine learning (ML) in Indian Ayurveda, it is clearly the case that there is no exercise of applying these advances within the Sri Lankan context. Specifically, existing literature is lacking in describing the architecture of AI-enabled personalized medicine recommendation systems capable of processing patient data, analyzing Prakriti (constitutional types), and returning evidence-based treatment recommendations. Due to the absence of such systems, Sri Lankan Ayurveda is significantly reliant on manual assessments and doctor expertise, limiting scalability.

Digital Integration and AI-Powered Recommendation Systems.

One of the biggest issues is the limited use of digital technology and artificial intelligence in Sri Lankan Ayurveda. While AI and machine learning have proven effective in predicting diseases and personalizing medicine in other medical fields, Ayurvedic medicine in Sri Lanka still lacks AI-based platforms that can provide personalized recommendations. There are no large-scale machine learning models trained on local patient data, which stops the development of context-specific treatment tools. Additionally, tools for automated Prakriti (body constitution) assessment are underdeveloped, making it hard to classify patients systematically and tailor treatment plans. Without these digital advancements, Ayurvedic care continues to rely on the subjective judgment of practitioners, leading to inconsistencies in diagnosis and treatment.

Integration with Modern Healthcare Systems

Ayurveda and modern medicine often coexist in Sri Lanka, and patients seek treatment from both the systems. However, no comprehensive scientific research on safe regimes of drug interactions between herbal drugs and modern medicines has been performed. This leaves a risk of harmful interactions unmonitored. There are also no treatment recommendation algorithms that combine both systems for optimum patient result. The lack of interoperability between Ayurvedic and traditional electronic health records also discourages efforts at providing holistic care. Lacking appropriate integration frameworks, Ayurveda stands alone and underutilized within the broader healthcare system as it has the potential to complement Western medical treatment.

Context-Specific Challenges

Most ML work in Ayurveda has been India-focused or generalized datasets, and Sri Lanka-specific solutions are lacking. Sri Lankan Ayurvedic medicine also uses locally available medicinal plants to a great degree and uses locally adapted treatment methods, yet no AI-based systems consider these variations. It is also a challenge to adapt Sanskrit-based Ayurvedic texts to a Sri Lankan medical context, as this requires translating classical concepts to local climate, diet, and cultural practices. Moreover, there has been scant research conducted on how regional differences affect treatment outcomes. Without this contextual adaptation, it is not possible to create precise, culturally relevant recommendation systems for Sri Lankan health requirements.

Technology-Enabled Accessibility

Access to Ayurvedic care in Sri Lanka is also restricted by the lack of technology-based solutions. Telemedicine and mobile health solutions, increasingly being used in modern healthcare to facilitate access levels, particularly in rural areas, are barely accessible for Ayurveda. There is little research on developing mobile applications through which patients can consult doctors, receive medicine prescriptions, or access health records electronically. Furthermore, digital health literacy among Ayurvedic practitioners is low, so even if such systems were developed, many would not be poised to adopt them. This digital divide limits the spread of Ayurveda and the collection of large-scale patient data that can fuel AI-driven innovation..

1.3 Research Problem

Although Ayurveda is popular in Sri Lanka, the system is still mostly not digital and hard to access for many communities, especially in rural areas. Patients in these regions often find it difficult to see qualified Ayurvedic practitioners and get trustworthy treatment advice. As a result, they may depend on informal guidance, self-treatment, or inconsistent practices, which can impact both the effectiveness and safety of care. This shows the urgent need for digital solutions to improve access to Ayurvedic healthcare.

One of the biggest hindrances in developing such solutions is the lack of sufficient data collected and computerized for Sri Lankan Ayurvedic practices. Information related to patient history, Prakriti classification, and medicinal plant usage is generally scattered, unrecorded, or maintained in non-computer formats. In the absence of structured data sets, it is challenging to build intelligent systems that generate effective and evidence-based recommendations for herbs. This disparity prevents Ayurveda from using contemporary AI and machine learning technology that would facilitate greater personalization and accuracy in treatment.

When we look at how herbal medicine recommendations are currently provided in Sri Lankan Ayurveda, there's a noticeable gap compared to how modern pharmacies and medical systems operate. While conventional medicine now uses digital platforms to suggest medications based on patient profiles, symptoms, and medical history, Ayurvedic herbal recommendations still depend entirely on direct consultations with traditional practitioners. This creates significant challenges for patients seeking appropriate herbal remedies, particularly those in rural communities where qualified herbalists are becoming increasingly rare. Traditional knowledge about which herbs work best for specific conditions, proper dosages, and individual constitutional considerations remains locked within the experience of aging practitioners. What's critically needed is a digital herbal recommendation system that can capture this traditional wisdom and make it accessible through technology. Such a platform could analyze a patient's symptoms, constitutional type, and health history to suggest appropriate herbal formulations, just as modern systems recommend pharmaceutical treatments. This represents our main research focus - creating an intelligent herbal recommendation platform that preserves traditional knowledge while making personalized herbal guidance available to Sri Lankan communities regardless of their geographic location or access to traditional practitioners.

2 OBJECTIVES

2.1 Main objective

The goal of this component is to collect and organize data from classical Ayurvedic texts to build a structured herb-disease dataset and then apply machine learning classification techniques to identify and recommend the most suitable herbs for a given disease considering multiple patient factors. This component suggests the best herbs linked to that disease while accounting for individual constitutional variations, allergic sensitivities, and presenting symptoms.

2.2 Specific objective

Data Collection & NER

Use Named Entity Recognition (NER) to extract herbs, diseases, doshas, symptoms, age factors, and contraindications from classical Ayurvedic texts

Dataset Creation

Organize the extracted data into a structured dataset linking diseases to herbs, along with features like dosha imbalance type, symptom profiles, age ,allergic contraindications, and severity levels

Model Training

Apply machine learning classification models (e.g., Random Forest, SVM, Decision Tree, Ensemble methods) to learn personalized herb recommendations based on disease diagnosis and multi-factor patient context.

Multi-Factor Integration

Develop algorithms that simultaneously consider disease type, dosha status, presenting symptoms, patient age, and known allergies for holistic recommendation generation.

Best Herb Recommendation

Predict and recommend the most suitable herb(s) for a disease-patient combination using the best-performing model based on accuracy, precision, and safety evaluation metrics.

3 METHODOLOGY

3.2 Component Diagram

The goal of the Medicine(Herb) Recommendation component is to find the best accurate recommendations model the component diagram outlines the workflow and structure broken down into 3 parts the Dataset Creation, Workflow ,Recommendation which outlines the steps the component Diagram works

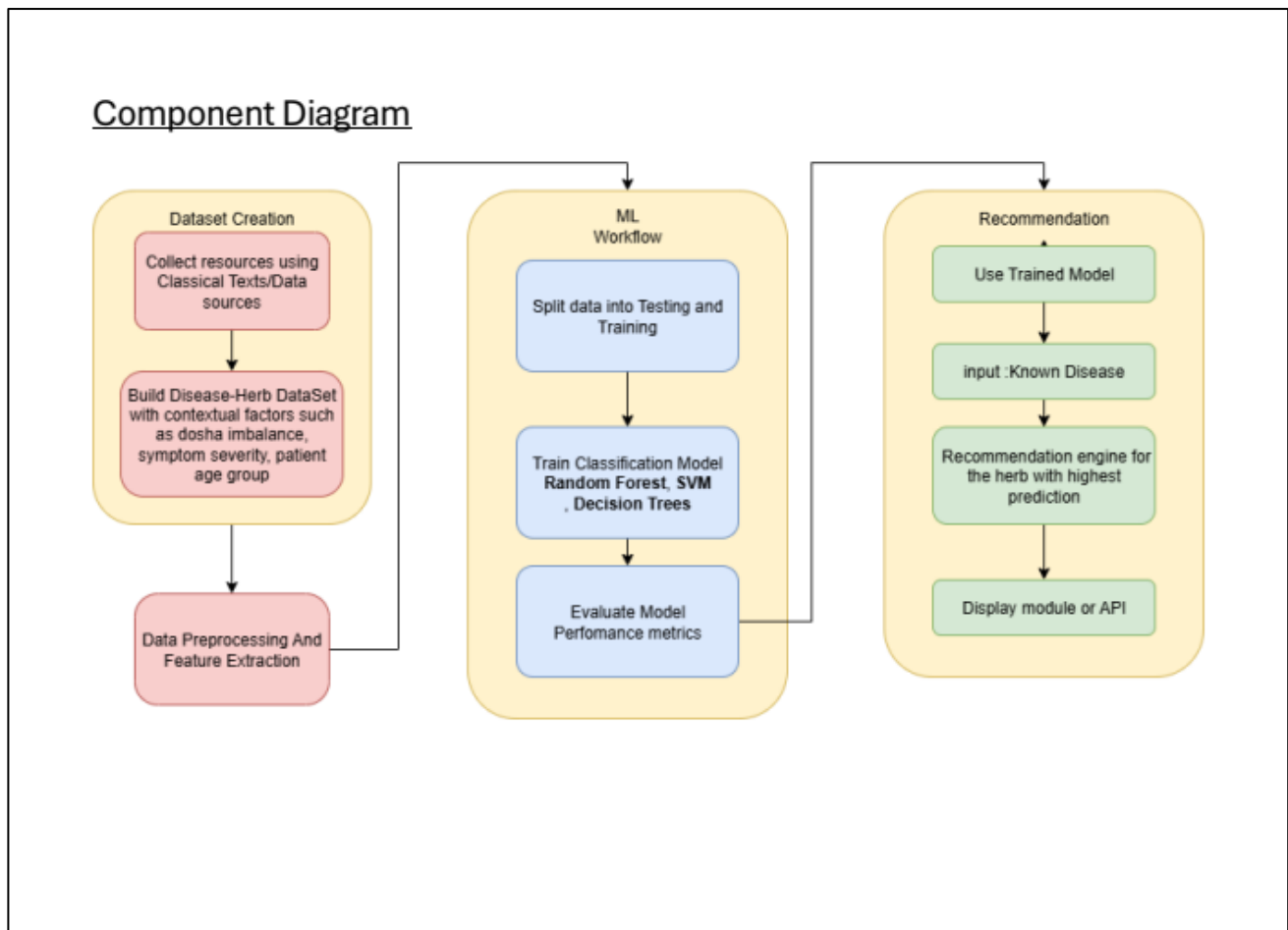


Figure 8 Component Diagram - Adaptive Question Generating Aptitude test

3.3 Tasks and Sub-Tasks

Task 1: Data Collection and preprocessing

Sub-task 1. 1: Digitize and process classical Ayurvedic texts

Sub-task 1. 2: Collect contemporary peer-reviewed research on Ayurvedic herb efficacy

Sub-task 1. 3: Gather clinical case studies and practitioner documentation

Sub-task 1. 4: Named Entity Recognition Implementation

- Develop custom NER models using spaCy or NLTK frameworks
- Train models to identify herbs ,diseases, symptoms, body constitutions
- Validate NER accuracy through review and verification

Task 2: Dataset Construction

Sub-task 2. 1: Create schema linking diseases, herbs, and contextual factors

Sub-task 2. 2: Feature Engineering

- categorical variables for age groups, gender, and constitutional types
- composite features representing herb-disease compatibility scores

Task 3: Model Development

Sub-task 3. 1: Algorithm Selection and Training

Sub-task 3. 2: Implement multiple classification algorithms:

- Random Forest (ensemble learning approach)
- Support Vector Machine (kernel-based classification)
- Decision Tree (interpretable rule-based model)

Sub-task 3. 3: Feature Selection and Optimization

- Apply feature importance analysis to identify key predictive variables

- Implement cross-validation techniques (k-fold, stratified sampling)
- Optimize hyperparameters using grid search and Bayesian optimization

Task 5: System Integration and Testing

Sub-task 5.1 :User Interface Development

- Design intuitive web-based interface for disease input and herb recommendation display
- Implement confidence scoring and alternative recommendation features
- Create comprehensive help documentation and usage guidelines

Sub-task 5.2 :Final Testing and Deployment

- Conduct end-to-end system testing
- Performance optimization and scalability testing

3.2 Requirement Gatherings

Data Gathering:

- Collection 1. Classical Ayurvedic Texts:
- Charaka Samhita (digitized version from National Institute of Ayurveda)
- . Local Ayurvedic Consultation Records (anonymized) – From Nawinna and Rajagiriya
- questionnaire answered by Ayurvedic practitioners.

4.5 Use Case

Table 2 Use Case Scenario Of the Adaptive Question Generating Aptitude Test

Use Case ID	01
Use case name	Receiving Personalized Herb Recommendation
Preconditions	The user is registered in the system. The user has entered their health-related details
Primary Actor	Patient
Main Scenario	1. The user logs in to the system. 2. The user Navigates to the medicine Section and enters the required details 3. The system analyzes the input using the stored knowledge base and model. 4. The system matches the symptoms with relevant herbs and treatments. 5. The system generates a list of recommended herbs, 6. The user can view the Most recommended 7. The user can download or view a simple report summarizing their recommendation.
Extensions	1.a If the user enters incomplete data, the system prompts them to complete the required fields. 1.b If a technical issue occurs, the system saves the user's progress and allows resumption later.

Post conditions	• The user's health input and recommendation history are saved for future reference. • The user receives a set of herbs best suited to their condition and constitution. • The recommendation can be used to guide further Ayurvedic consultation or personalized activities.
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5 CONCLUSION

This review shows how the timeless wisdom of Ayurveda can thrive with modern artificial intelligence. The evidence indicates that machine learning, especially models like Random Forest, can make Ayurvedic knowledge more accessible, accurate, and relevant without losing its holistic roots. With accuracy rates as high as 95% and strong public interest, it is clear that people are ready for digital tools that can guide them toward personalized herbal care and balanced living. At the same time, the findings remind us that technology alone is not enough. To truly help patients, especially those in rural areas, these systems must be based on diverse and reliable datasets, tested against real-world healthcare needs, and validated through strict clinical trials. By combining Ayurveda's human-centered philosophy with the precision of AI, we have a real chance to create solutions that are effective, trustworthy, and accessible to everyone.

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